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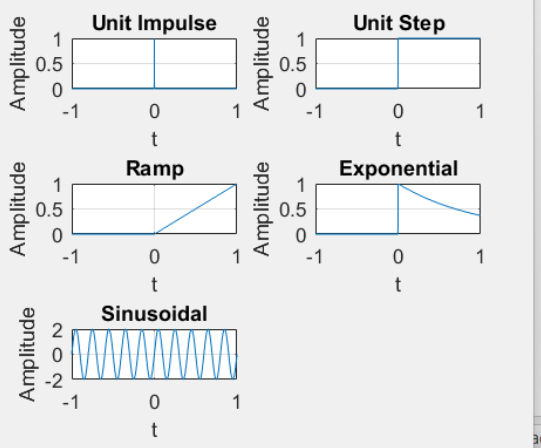
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Figure 1

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SOS1.m

```
% Experiment 1: Generation of Continuous-Time Signals
clc; clear; close all;
t = -1:0.001:1;
A = 2; f0 = 5; a = 1;
imp = double(abs(t)<0.0005);
step = double(t>=0);
ramp = t.*step;
expSig = exp(-a*t).*step;
sinSig = A*sin(2*pi*f0*t);
figure;
subplot(3,2,1); plot(t,imp); grid on; title('Unit Impulse'); xlabel('t'); ylabel('Amplitude');
subplot(3,2,2); plot(t,step); grid on; title('Unit Step'); xlabel('t'); ylabel('Amplitude');
subplot(3,2,3); plot(t,ramp); grid on; title('Ramp'); xlabel('t'); ylabel('Amplitude');
subplot(3,2,4); plot(t,expSig); grid on; title('Exponential'); xlabel('t'); ylabel('Amplitude');
subplot(3,2,5); plot(t,sinSig); grid on; title('Sinusoidal'); xlabel('t'); ylabel('Amplitude');
```

Command Window

Name	Value
a	1
A	2
expSig	*1x2001 double*

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5   imp = double(abs(t)<0.0005);
6   step = double(t>=0);
7   ramp = t.*step;
8   expSig = exp(-a*t).*step;
9   sinSig = A*sin(2*pi*f0*t);
10  figure;
11  subplot(3,2,1); plot(t,imp); grid on; title('Unit Impulse'); xlabel('t'); ylabel('Amplitude');
12  subplot(3,2,2); plot(t,step); grid on; title('Unit Step'); xlabel('t'); ylabel('Amplitude');
13  subplot(3,2,3); plot(t,ramp); grid on; title('Ramp'); xlabel('t'); ylabel('Amplitude');
14  subplot(3,2,4); plot(t,expSig); grid on; title('Exponential'); xlabel('t'); ylabel('Amplitude');
15  subplot(3,2,5); plot(t,sinSig); grid on; title('Sinusoidal'); xlabel('t'); ylabel('Amplitude');
```

Command Window

Workspace

Name	Value
a	1
A	2
expSig	*1x2001 double*

Ln 15 Col 2

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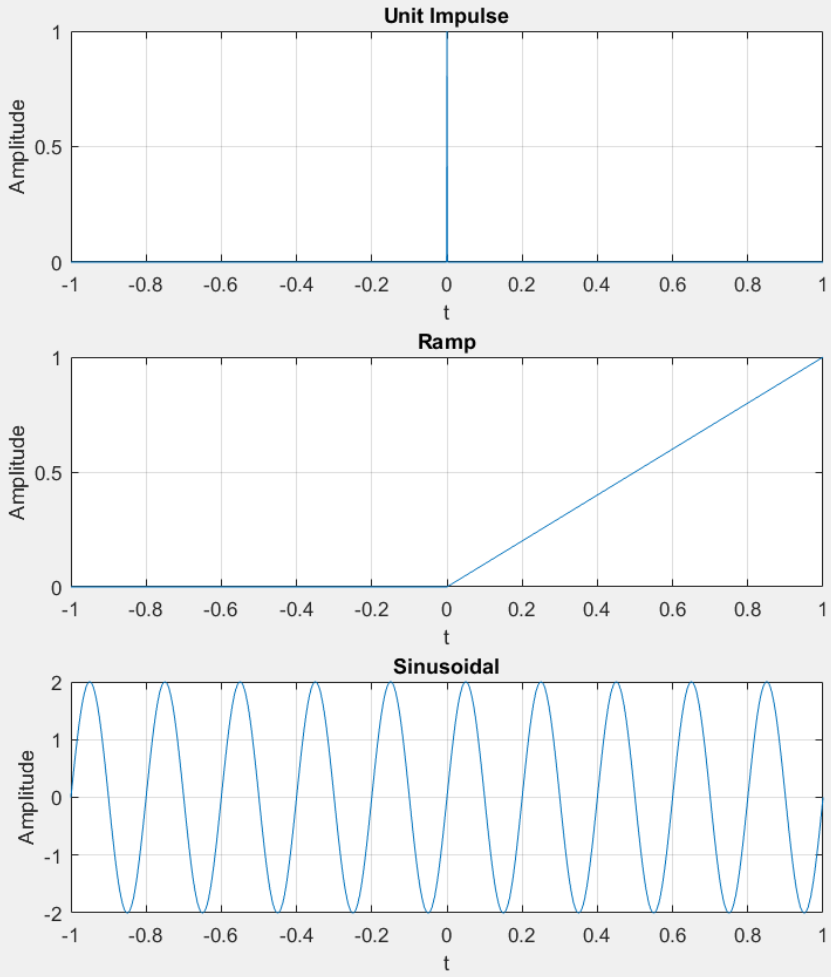
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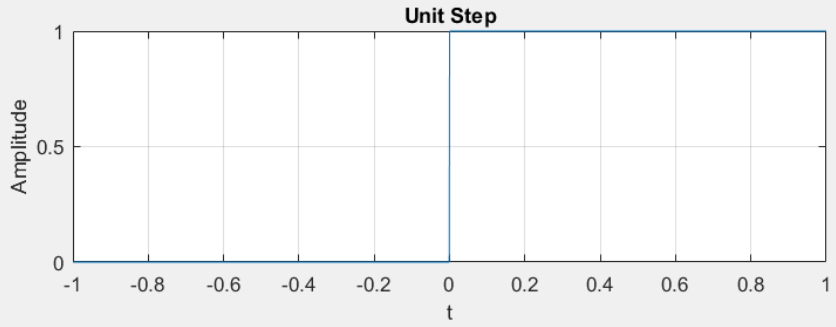
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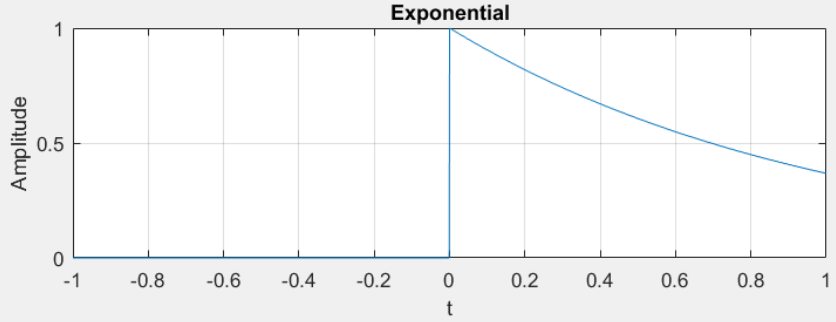
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Figure 1

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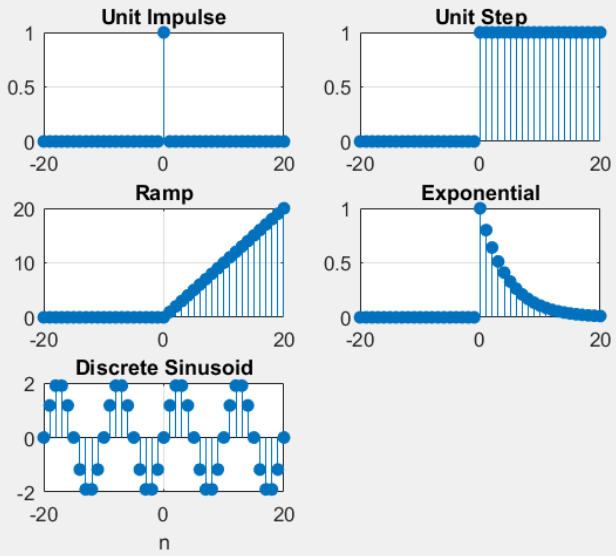
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SOS1.m | SOS2.m

```
1   % Experiment 2: Generation of Discrete-Time Signals
2   clc; clear; close all;
3   n = -20:20;
4   A = 2; f0 = 0.1; a = 0.1;
5   imp = double(n==0);
6   step = double(n>=0);
7   ramp = n.*step;
8   expSig = (0.8).^n .* step;
9   sinSig = A*sin(2*pi*f0*n);
10  figure;
11  subplot(3,2,1); stem(n,imp,'filled'); grid on; title('Unit Impulse');
12  subplot(3,2,2); stem(n,step,'filled'); grid on; title('Unit Step');
13  subplot(3,2,3); stem(n,ramp,'filled'); grid on; title('Ramp');
14  subplot(3,2,4); stem(n,expSig,'filled'); grid on; title('Exponential');
15  subplot(3,2,5); stem(n,sinSig,'filled'); grid on; title('Discrete Sinusoid');
16  xlabel('n');
```

Figure 1

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Workspace

Name	Value
a	0.1000
A	2
expSig	*1x41 double*

Command Window

Ln 16 Col 1

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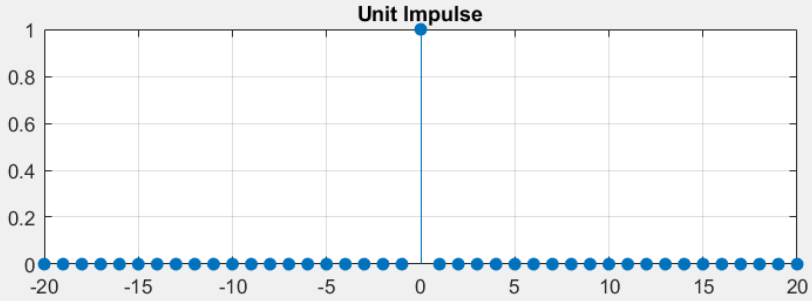
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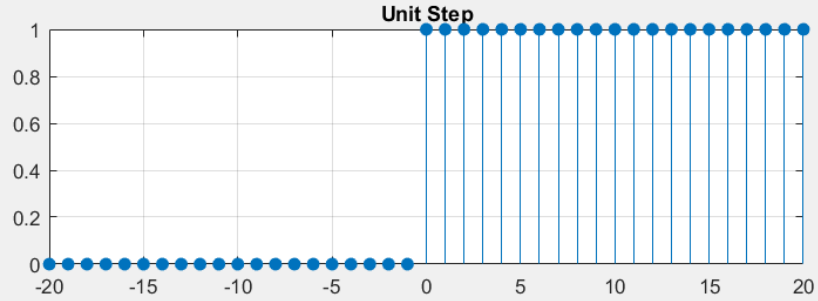
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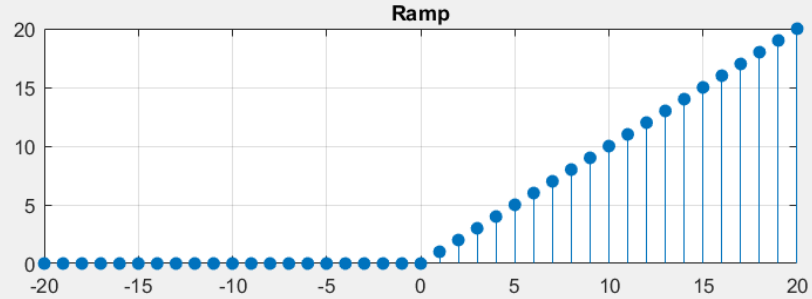
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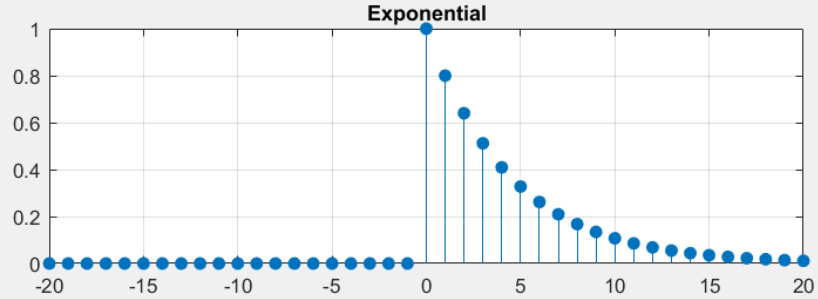
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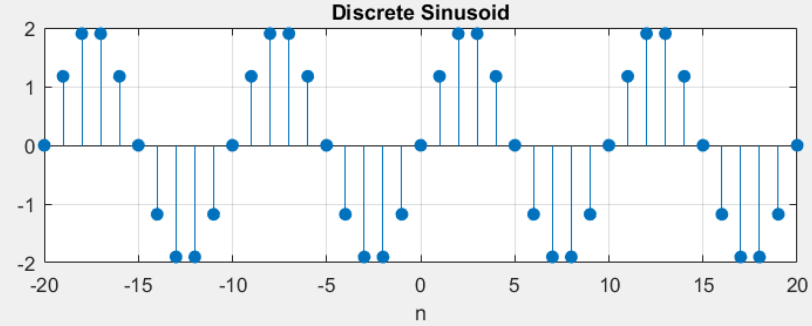
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SOS1.m | SOS2.m

```
1   % Experiment 2: Generation of Discrete-Time Signals
2   clc; clear; close all;
3   n = -20:20;
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5   imp = double(n==0);
6   step = double(n>=0);
7   ramp = n.*step;
8   expSig = (0.8).^n .* step;
9   sinSig = A*sin(2*pi*f0*n);
10  figure;
11  subplot(3,2,1); stem(n,imp,'filled'); grid on; title('Unit Impulse');
12  subplot(3,2,2); stem(n,step,'filled'); grid on; title('Unit Step');
13  subplot(3,2,3); stem(n,ramp,'filled'); grid on; title('Ramp');
14  subplot(3,2,4); stem(n,expSig,'filled'); grid on; title('Exponential');
15  subplot(3,2,5); stem(n,sinSig,'filled'); grid on; title('Discrete Sinusoid');
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Workspace

Name	Value
a	0.1000
A	2
expSig	*1x41 double*

Command Window

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